



The Influence of Working from Home vs. Working at the Office on Job Performance in a Hybrid Work Arrangement: A Diary Study

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Abstract

Working from home (WFH) has become increasingly common, and its impact on job performance is particularly interesting to researchers. Prior research has established a positive correlation between remote work and job performance, but the underlying mechanisms still need to be fully understood. This diary study addresses this issue by applying the Job Demands-Resources (JD-R) theory and developing a mediation model that identifies two processes responsible for the positive and negative relationships of WFH (compared to working at the office or WATO) with job performance. The study was conducted in a public organization, and 203 employees engaged in hybrid work arrangements answered an online survey for eight consecutive workdays. Results showed that working from home (WFH) had an indirect positive relationship with job performance through daily concentration and work engagement. Instead, no significant indirect relationship of WFH with daily job performance through daily social isolation and tension was observed. However, a disengagement “effect” was found, as the indirect relationship of WFH with daily job performance via daily social isolation and work engagement was negative and statistically significant. Theoretically, this study supports a motivational mechanism linking WFH to job performance, does not support the health-impairment process triggered by WFH, and emphasizes the importance of considering the complex interplay between job demands and motivational states. From a practical standpoint, the study suggests that organizations implementing hybrid work should enhance employee motivation, for instance, by providing tools to improve employee concentration and addressing potential feelings of social isolation when working from home.

Keywords Working from home · Job performance · Job Demands-Resources theory · Hybrid work · Diary study

The COVID-19 pandemic has brought about a seismic shift in the way people work, with remote work or working from home (WFH) becoming increasingly prevalent. In the USA, only 5.7% of workers primarily worked from home in 2019 (Ortiz, 2023). By 2022, this figure had risen to 22%, with a further 17% adopting a hybrid model that combined office work with remote work (Statista Research Department, 2023). The pandemic has also had a significant impact on European workers, with almost 40% of employees in the European Union (EU-27) switching to

full-time teleworking during the outbreak (Eurofound, 2021). Although the number of European workers exclusively working from home declined in 2021, hybrid work, where employees alternate between working at the firm’s facilities and at home, has become increasingly common. In 2021, 49% of Europeans stated that they would like to work from home several days a week or a month, indicating that remote work is likely to remain a popular option for many workers (Eurofound, 2021). As Forbes suggested, remote work is “here to stay” in the post-pandemic era, and hybrid work is likely to be and become the new normal for many employees (Henley, 2021).

Previous studies have addressed the relationship between working in remote settings (also called telecommuting or telework; Toscano & Zappalà, 2020a) and employee performance. In two valuable meta-analyses, Gajendran & Harrison (2007) and Martin & MacDonnell (2012) found positive relationships between working in remote settings and job performance. Specifically, Gajendran & Harrison (2007) found that the average uncorrected correlation between telecommuting and job

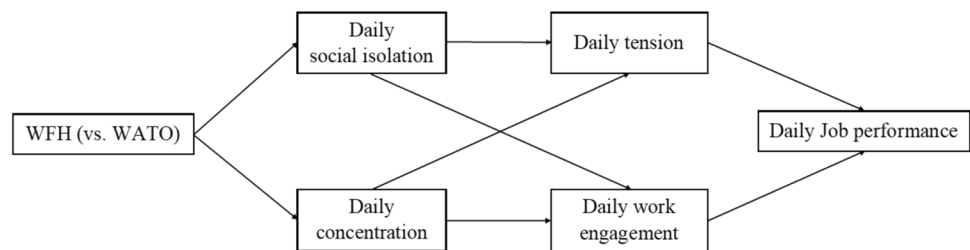
Additional supplementary materials may be found here by searching on article title <https://osf.io/collections/jbp/discover>

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Fig. 1 Research model. Note. WFH, working from home; WATO, working at the office



performance (measured with supervisor ratings or archival records) was 0.18 (95% CI = [0.09; 0.26]; the correlation corrected for unreliability was 0.19). Moreover, Martin & MacDonnell (2012) found that the weighted mean correlation corrected for unreliability between telework and subjective ratings of performance was 0.16 (95% CI = [0.09; 0.23]). However, in most of the empirical studies, telework was operationalized as the amount of scheduled time (e.g., two days a week) spent working away from a central work location (e.g., the office; Gajendran & Harrison, 2007; Golden & Gajendran, 2019). Using a different research design, researchers in two studies (Angelici & Profeta, 2023; Bloom et al., 2015) randomly assigned participants to telework vs. non-telework groups and observed a positive association between WFH (vs. working at the office-WATO) and job performance. This means that previous studies rarely distinguished between the impact of WFH vs. WATO on job performance on a daily basis and used within-subject designs. To the best of our knowledge, only two studies (Delanoeije & Verbruggen, 2020; Vega et al., 2015) did so. They also found that WFH (as compared to WATO) enhanced job performance. However, none of the two aforementioned studies examined the mediators that might explain the relationship between these two variables. This omission is problematic because it shows that we do not fully understand why WFH (as compared to WATO) enhances job performance. Fixing this problem is important for both theoretical and practical reasons. From a theoretical perspective, uncovering the mechanisms that explain why two variables are related helps move knowledge forward given that it provides a deeper understanding of the addressed relationship (Preacher & Hayes, 2004). Moreover, after reviewing the telecommuting literature, Allen and colleagues (2015) asserted that the relationship between WFH and job performance “is complex, with the potential for simultaneous benefits and drawbacks” (p. 60). This underlines how crucial it is to identify and distinguish the mediators that foster the benefits of WFH from those mediators that potentially trigger its drawbacks. Furthermore, considering the psychological processes occurring on a daily basis within the same individuals rather than between different ones, allows for a more fine-grained knowledge about the investigated relationship. From a practical perspective, identifying these different mediators enables us to suggest ways to promote the functional influence of WFH on job performance and mitigate its dysfunctional influence.

Thus, the goal of our diary study is to test a mediation model to ascertain some of the mediators that transmit the positive and the negative influences of WFH (vs. WATO) on job performance. Our research model is displayed in Fig. 1. Based on the Job Demands-Resources (JD-R) Model (Bakker & Demerouti, 2017; Demerouti et al., 2001), our model posits that WFH (vs. WATO) is related to daily job performance through both a health-impairment process and a motivational process. According to the health-impairment process, WFH is positively related to employees’ daily social isolation, which in turn is positively related to daily job tension, which in turn is negatively related to daily job performance. Thus, WFH (vs. WATO) has an indirect negative effect on daily job performance. According to the motivational process, WFH is positively related to employees’ daily concentration, which in turn is positively related to daily work engagement, which in turn is positively related to daily job performance. Thus, WFH (vs. WATO) also has an indirect positive effect on daily job performance. Based on previous meta-analysis (Alarcon, 2011; Crawford et al., 2010; Lee & Ashforth, 1996) and appropriate theoretical accounts, cross-links between the two processes are also expected.

Our study intends to make several contributions to literature. First, we try to identify some of the mechanisms through which WFH is indirectly related to daily job performance. The identification of the processes through which two variables are related “is often an increase in knowledge and an important refinement of the theory” (Spencer et al., 2005, p. 398). By investigating the hypothesized mechanisms, we contribute to a deeper understanding of the relationship between WFH (vs. WATO) and daily job performance, advancing knowledge in the field and moving our discipline forward (Mathieu et al., 2008). Second, we propose that the relationship between WFH (vs. WATO) and daily work performance is complex and posit that it includes both positive and negative indirect relationships. We provide a theoretical explanation for these relationships that helps us better understand why WFH (vs. WATO) is a two-sided phenomenon, which should preclude the adoption of simpler approaches focused solely on its advantages or limitations. Third, based on our theoretical arguments and previous empirical evidence (e.g., Alarcon, 2011; González-Romá et al., 2020), we propose that the JD-R theory should

be extended to include cross-links between the motivational and health-impairment processes embedded within this theory. This extension can improve the capacity of the JD-R theory to explain complex phenomena. Finally, we provide organizations and Human Resources (HR) professionals with ideas to guide the adoption of hybrid work arrangements, maximize its benefits, and limit its shortcomings. Considering that hybrid work will represent the new way of working for many employees, these ideas may have relevant consequences for employees and organizations.

Theoretical Framework and Hypotheses

The Job Demands-Resources Model

This study is theoretically based on the Job Demands-Resources (JD-R) theory (Bakker & Demerouti, 2017; Demerouti et al., 2001). This theory posits that job characteristics can be categorized into job demands and job resources. Job demands refer to physical, psychological, and contextual aspects of the work that lead to energy depletion and cause employee energy loss and stress. Job demands trigger a health-impairment process that leads to decreased well-being and performance. On the other side, job resources are physical, psychological, and contextual aspects of the job that are functional in achieving work goals and stimulate personal growth, learning, and development (Bakker & Demerouti, 2017). Job resources trigger a motivational process that leads to increased well-being and performance.

The JD-R theory is a broad approach applicable to many job characteristics and outcomes (Schaufeli & Taris, 2014). In this study, the JD-R theory provides a helpful lens for examining the potential negative and positive relationships of WFH with the job performance of employees that adopt a hybrid mode of work. Our model posits that WFH is negatively related to job performance via employees' social isolation, considered a contextual job demand, and tension, considered a strain response. We consider employee social isolation as a job demand based on the following reasons. Social isolation constitutes the perception of a decrease or absence of social interactions with colleagues. Working from home implies being physically distant from colleagues. According to the Self-Determination Theory (Ryan & Deci, 2000), physical distance can also imply a psychological distance that can hinder the fulfillment of the need for relatedness. To prevent this, employees working from home may invest energy in connecting and interacting with their colleagues via electronic media. This additional effort can contribute to depleting employees' energy reservoir, which will lead to increased strain (e.g., tension). If, despite this additional effort, employees working from home cannot satisfy

their need for relatedness, they will experience frustration, which in turn will generate tension (Chen et al., 2015).

Our model also posits that WFH is positively related to job performance via employees' job concentration (a job resource), and work engagement (a positive motivational state) (Bakker et al., 2014). We define job concentration as the employees' state of focusing mental resources (i.e., attention) toward work tasks (Nohe et al., 2014). We consider job concentration as a job resource because it increases employees' capacity to deal with their job activities and achieve their work goals. This conceptualization is consistent with ten Brummelhuis & Bakker's (2012) classification of focus and attention as examples of psychological job resources. According to the Self-Determination Theory (Ryan & Deci, 2000), job concentration enables the fulfillment of employees' need for competence and thus stimulates motivational states such as work engagement (Biron & van Veldhoven, 2016; Ryan & Deci, 2000), which in turn is positively associated with job performance (Bakker et al., 2014).

WFH and Job Performance through a Health-Impairment Process

In this study, we posit that, on days they work from home, employees engaged in a hybrid work program experience more social isolation than on days they work on-site. Working from home implies a physical and psychological distance from colleagues (Ryan & Deci, 2000), which contributes to increasing their perceptions of social isolation. The Need to Belong theory suggests that people need frequent personal interactions with others to feel cognitively and emotionally well (Baumeister & Leary, 1995). Similarly, the Self-Determination Theory suggests that, to be motivated, people need to experience a sense of belonging and connection with others (Ryan & Deci, 2000). Although, thanks to digitalization, social interactions are not precluded to workers on the days they work at home, it is likely that the social exchanges they experience with colleagues on those days are of lower quality than on the days they work on-site (Fonner & Roloff, 2010). The absence of face-to-face relationships contributes to employees' social isolation on days they work from home.

The scientific literature has long considered employee isolation as one of the most notable drawbacks of remote working arrangements. Twenty years ago, Cooper & Kurland (2002) already cited isolation as a telework-related demand both at the professional level (fewer opportunities for promotions and organizational rewards) and at the social level. Remote working implies losing informal interactions with colleagues to a great extent. Previous studies (Mann & Holdsworth, 2003; Pyöriä, 2011) have supported the role of WFH on the generation of the perception of being cut off from regular social networks, and loneliness.

We also posit that daily social isolation is positively related to the strain variable of tension (i.e., the feeling of restlessness and anxiety felt by a person at a given time). We focused on tension because it is one of the main consequences of social isolation (Robb et al., 2020) and, according to the JD-R theory (Bakker & Demerouti, 2017), it is one of the strain variables linking job demands and job performance.

The JD-R theory posits that job demands are related to psychological costs. In this framework, social isolation represents a hindrance job demand (LePine et al., 2005), since it is, according to the JD-R theory, a contextual aspect of the work that interferes with the employees' capacity to meet their human need for relatedness. Failure to meet this need generates a need frustration, which in turn generates tension (Chen et al., 2015). Previous studies support that social isolation is related to strain responses (Galanti et al., 2021; Toscano & Zappalà, 2020b).

Finally, following the health-impairment process, our model posits that tension is negatively related to daily job performance. The JD-R model underlines the negative relationship between strain and job performance. According to this relationship, tension leads to a depletion of energy, which restrains employees' performance. Tension clouds humans' coping possibilities, worsens normal cognitive functions, and leads employees to feel less able to complete tasks at work, and thus perform worse (Dunk, 1993; Halbesleben & Bowler, 2007; Zhang et al., 2022).

Considering all the arguments presented above, we propose the following hypothesis:

Hypothesis 1: There is a negative indirect effect of WFH (vs. WATO) on daily performance sequentially mediated by daily social isolation and daily tension, so that on days in which employees work from home (as compared to days in which they work at the office) they experience more social isolation, which in turn is positively related to daily tension, which in turn is negatively related to daily performance.

WFH and Job Performance through a Motivational Process

We posit that, on days they work from home, employees engaged in a hybrid work arrangement experience more job concentration than on days they work at the office. According to the Self-Determination Theory, people try to satisfy their need for competence, that is, to feel and experience that they are effective in dealing with their environment (Ryan & Deci, 2000). This implies that, when working, employees leverage personal and environmental resources to accomplish their work at best to feel competent. A crucial psychological job resource that enables good job performance is

job concentration (Nohe et al., 2014). Job concentration is fostered by choosing home as a place to work. Working from home helps to carry out work activities in less noisy contexts, with fewer interruptions and potentially better opportunities to focus attention on work tasks (Xiao et al., 2021).

Previous research reported concentration as a focal advantage of remote work. For example, Van Sell & Jacobs (1994), almost 30 years ago, found that employees were less distracted at home than in the office and that managers reported that workers took work home to avoid being interrupted in the office. More recent studies have empirically confirmed the relationship between WFH and job concentration, either through cross-sectional surveys (Vittersø et al., 2003) or a diary study (Biron & van Veldhoven, 2016). Additionally, some studies have positioned job concentration as the factor that explains the job performance improvement in teleworking environments compared to traditional office settings (Gajendran & Harrison, 2007). Thus, we posit that, when employees work from home (as compared to working at the office), they experience more job concentration.

We also posit that daily job concentration is positively related to daily work engagement, which is a "positive, fulfilling, work-related state of mind that is characterized by vigor, dedication, and absorption" (Schaufeli et al., 2002, p. 74). We focus on work engagement because, according to the JD-R theory (Bakker & Demerouti, 2017), it is the motivation construct that best links job resources and job performance through a motivational process.

The JD-R theory argues that job resources, like job concentration, are positively related to motivational states, such as work engagement. Focusing on the specific job resource included in our model, job concentration helps employees carry out their work activities and achieve their goals by focusing their cognitive resources (attention) on the job tasks. Thus, job concentration allows employees to feel they can perform their job efficiently. Therefore, according to the Self-Determination Theory, job concentration contributes to fulfilling employees' need for competence, which in turn fosters motivation states like work engagement (Ryan & Deci, 2000).

Finally, along the motivational process, our model posits that daily work engagement is positively related to daily job performance. This relationship is supported by the JD-R model and empirical evidence (Bakker et al., 2014). Work engagement widens the person's thought and action repertoire and enables resources that lead to an enhancement of job performance (Fredrickson, 2001). It triggers enhanced cognitive functioning, cognitive openness, and behavioral flexibility (Reijseger et al., 2010), which lead to better job performance (Bond & Flaxman, 2013; Duff et al., 2012).

Therefore, we argue that, in parallel to the health-impairment process, a motivational process also relates WFH to job performance. On days they work from home, employees in

a hybrid work arrangement experience a higher level of job concentration (compared to days in which they work at the office), which in turn leads to a higher level of daily work engagement, which in turn leads to a higher level of daily performance. Therefore, we hypothesize:

Hypothesis 2: There is a positive indirect effect of WFH (vs. WATO) on daily job performance sequentially mediated by daily concentration and daily work engagement, so that on days in which employees work from home (as compared to days in which they work at the office) they experience more concentration, which in turn is positively related to daily work engagement, which in turn is positively related to daily performance.

WFH and Job Performance through Cross-Relationships

Although the health-impairment and motivational processes constitute two different pathways, it is necessary to consider that their effects may be intertwined. Indeed, cross-links between job resources and job demands, on the one hand, and functional and dysfunctional mediators (e.g., work engagement and job tension, respectively) have been reported in the JD-R literature. Some studies (Bakker et al., 2003; Schaufeli & Bakker, 2004) and meta-analyses (Alarcon, 2011; Crawford et al., 2010; Lee & Ashforth, 1996) found that job resources are negatively related to employee strain, whereas job demands are negatively related to motivational states. Thus, we include these cross-links in our mediational model, so that we expect a negative relationship between social isolation and work engagement, and between job concentration and job tension.

Social isolation is negatively related to work engagement (Galanti et al., 2021; Vansteenkiste et al., 2020). In a work context, experiencing social isolation means feeling less connected to colleagues than one would like. The experience of social isolation hampers the satisfaction of a primary human need, that of relatedness, which is one of the bases of motivation (Deci et al., 2017; Ryan & Deci, 2000). As a result, motivational states such as work engagement are adversely affected. Thus, we posit that WFH is negatively related to job performance through the sequential mediation of social isolation and work engagement. Therefore, considering the arguments presented above, and the hypothesized negative relationship between WFH and social isolation as well as the hypothesized positive relationship between work engagement and job performance, we hypothesize the following:

Hypothesis 3: There is a negative indirect effect of WFH (vs. WATO) on daily performance sequentially mediated by daily isolation and daily work engagement, so that on

days in which employees work from home (as compared to days in which they work at the office), they experience more isolation, which in turn is negatively related to daily work engagement, which in turn is positively related to daily performance.

We also posit that daily concentration is negatively related to daily tension. This relationship is based on self-regulatory mechanisms (Hockey, 1993; Robert & Hockey, 1997). On days in which employees experience high levels of concentration, they can focus their energy and attention on the tasks that need to be performed. This focused attention avoids potential distractors and the conflict between their associated courses of action and the goals of the employee's tasks. Under this non-conflict condition, employees will experience low levels of tension. On the contrary, on days in which employees experience low levels of concentration on their own tasks and distractors catch their attention, conflicts between their tasks' goals and other courses of action triggered by distractors will arise. Under this condition, employees will experience higher levels of tension. Thus, considering the arguments presented above, and the hypothesized positive relationship between WFH and concentration, and the hypothesized negative relationship between job tension and job performance, we hypothesize the following:

Hypothesis 4: There is a positive indirect effect of WFH (vs. WATO) on daily performance sequentially mediated by daily concentration and daily tension, so that on days in which employees work from home (as compared to days in which they work at the office), they experience more concentration, which in turn is negatively related to daily tension, which in turn is negatively related to daily performance.

Finally, because previous research has reported empirical evidence supporting a direct relationship between WFH (vs. WATO), on the one hand, and work engagement (Delanoeije & Verbruggen, 2020) and job tension (Islam et al., 2023; Wang et al., 2023), on the other, we controlled for these relationships when estimating the complex indirect relationships involved in our research model.

Method

Procedure and Sample

This diary study was conducted through a collaboration between the researchers and an Italian municipality, which invited its hybrid workers to voluntarily participate in the study after providing them with information about the initiative. Participants were surveyed for eight consecutive

working days, with five days in the first week (Monday through Friday) and three days in the second week (Monday through Wednesday). We were not able to survey participants on Thursday and Friday of the second week due to local holidays. The data was collected through an online platform (Qualtrics) in June 2021.

The hybrid workers received invitation emails that were prepared by the researchers but agreed upon with, and sent by, the organization's HR department. Employees were expected to work from 8:00 am to 2:00 pm on Monday, Wednesday, and Friday, and from 8:00 am to 5:00 pm, with a 1-h lunch break, on Tuesday and Thursday. On shorter workdays, participants received the link to the questionnaire and an instruction to complete it at 1:00 pm. On longer workdays, participants received a first email at 1:00 pm with a link to the questionnaire and an instruction to complete it just before the end of the workday. A second email was sent on these longer days at 4:30 pm in which participants were instructed to complete the questionnaire immediately.

Our sample consists of 203 civil servants with varied occupations, including administrative staff, accountants, architects, engineers, local police officers, librarians, and educators. These employees were engaged in hybrid work activities within the municipality. Each employee could spend 1 or 2 days per week working from home. The specific days on which employees worked from home were not determined by organizational policies, but decided in agreement with the corresponding supervisor on a weekly basis. In total, we collected 751 valid daily responses (503 referring to workdays spent at the office and 248 workdays spent at home), for an average of 3.70 days per participant (response rate = 46.2%). The participants had an average age of 48.5 years ($SD = 9.75$; $min = 24$; $max = 66$), and 74.3% of them were women. In terms of education, 58.3% had a college degree, 40% had a high school diploma, and the rest had lower levels. Their tenure averaged 16.13 years ($SD = 11.97$; $min = 0$; $max = 43$).

All participants provided informed consent before participating in each daily data collection. This study was approved by the Ethical Committee of the corresponding author's university. The study was designed and carried out following ethical guidelines and ensuring the protection of the participants' rights.

Measures

Whenever it was possible, we used measures previously validated in Italian samples. In other cases, we followed Brislin's (1986) back-translation procedure: the original English versions of the measures were translated into Italian by a bilingual researcher and then back-translated into English by another bilingual researcher. Discrepancies between the original and the back-translated versions were discussed and

solved by the involved researchers. Moreover, to guarantee accuracy, a third bilingual researcher checked the whole process and its outcomes. In the following, we describe the measures used in the study; the full list of items is mentioned in the Appendix.

WFH (vs. WATO)

Working from home (vs. working at the office) was measured through the following question: "Today, have you worked ...?". The response options were as follows: "0. At the office"; "1. From home"; "Somewhere else. Please specify." The last option was selected only ten times by nine different participants. These responses were associated with days on which the employees extended their workday and worked both from home and at the office, or the employees (e.g., construction technicians) visited construction sites. Given that our study focused on examining the indirect relationship between WFH vs. WATO and performance, and considering the small number of respondents who answered "Somewhere else" and the heterogeneity of places involved, we removed these cases from our database.

Daily Social Isolation

This variable was measured with a 4-item version of the scale elaborated by Golden et al. (2008). This short version was previously validated in Italian samples (Galanti et al., 2021; Toscano & Zappalà, 2020b). We adapted it to a daily context. An example of item was: "Today, I felt isolated". The response scale was a graded scale that varied from 1. "Not at all" to 7. "Very much". We submitted the hypothesized one-factor model to a multilevel Confirmatory Factor Analysis (CFA) using Mplus (Muthén & Muthén, 2017). To assess the fit of multilevel Structural Equation Modeling (SEM) models, different scholars recommend using level-specific fit indices because they allow researchers to examine fit at the between-person (Level 2) and within-person (Level 1) levels of analyses separately (González-Romá & Hernández, 2017, 2022; Ryu, 2014). Moreover, this approach avoids using overall fit measures that do not tell where potential fit problems may exist. Mplus computes the Standardized Root Mean Square Residual (SRMR; an absolute fit index) both for the within and between parts of multilevel SEM models. The values obtained for the hypothesized one-factor model were good (SRMR-within = 0.032; SRMR-between = 0.042). All the factor loadings were statistically significant ($p < 0.001$). We used the values obtained for the factor loadings and the item error variances to compute the omega reliability coefficient for the within part of the model (omega-within; Bolger & Laurenceau, 2013). This coefficient estimates the precision of the scale to measure

within-person changes over time. The omega-within value obtained for the social isolation scale was 0.80.

Daily Tension

This variable was measured using a differential semantic item from the scale by Wilhelm & Schoebi (2007), adapted to the daily context. The item was: “At this moment, I feel: Relaxed 1. ... 2 ... 3 ... 4 ... 5 ... 6 ... 7. Tense”. In a sample composed of 1150 university students (77% women; mean age = 22.3 years, SD = 5.2), the correlation between this item and Segura & González-Romá’s (2003) 6-item scale for measuring the affective well-being dimension of tension-calm was 0.81. This result supports the validity of our measure.

Daily Job Concentration

This variable was measured with a single item from the diary scale by Nohe et al. (2014): “Today, I had total concentration while working”. The response scale was a graded scale that varied from 1. “Not at all” to 7. “Very much”. In a previous study (Herrando et al., 2018), this same item was administered as part of a 3-item scale that showed a Cronbach’s alpha of 0.895. Based on this value, and applying the Spearman-Brown formula that relates test length and reliability (de Vet et al., 2017), we estimated the reliability of our single item, which equaled 0.74.

Daily Work Engagement

This was measured with the 3-item version (UWES-3, Schaufeli et al., 2019) of the UWES scale (Schaufeli et al., 2002, 2006), adapted to a daily context. An example item was: “Today, I felt immersed in my job.” The response scale was a graded scale that varied from 1. “Not at all” to 7. “Very much.” We submitted the hypothesized one-factor model to a multilevel CFA. Because the model was just identified, the model’s fit was perfect (SRMR-within = 0.000; SRMR-between = 0.000). All the factor loadings were statistically significant ($p < 0.001$). The reliability coefficient omega-within was 0.72.

Daily Job Performance

This variable was measured with the four-item scale elaborated by Pettit et al. (1997) adapted to a daily context. An example item was: “Today, how do you think your supervisor would evaluate the quality of your performance?”. The response scale was a graded scale that varied from 1. “Very poor” to 7. “Excellent.” There is empirical evidence supporting the validity of this measure. Pettit and colleagues (1997) reported positive correlations between

employees’ and supervisors’ scores on the scale items focused on performance quality ($r = 0.32$, $p < 0.01$) and performance quantity ($r = 0.19$, $p < 0.01$). Gürbüz et al. (2023) reported positive time-lag correlations between work engagement and successive job performance (r s with a 2- and 4-month time-lag were 0.41, $p < 0.01$, and 0.32, $p < 0.01$, respectively). Finally, Kooij et al. (2020), in a diary study across five consecutive working days, reported a within-subject correlation between daily work engagement and daily job performance of 0.40 ($p < 0.01$). Moreover, we submitted the hypothesized one-factor model to a multilevel CFA. The model’s fit was adequate (SRMR-within = 0.055; SRMR-between = 0.085). All the factor loadings were statistically significant ($p < 0.001$). The reliability coefficient omega-within was 0.88.

To examine the discriminant validity of the scales used to measure daily social isolation, daily work engagement, and daily job performance, we submitted them to a multilevel CFA. The fit of the 3-factor model was acceptable (SRMR-within = 0.051; SRMR-between = 0.094). We compared the fit of this model with the fit of a one-factor model. The latter model showed a bad fit to data (SRMR-within = 0.299; SRMR-between = 0.391). Under the 3-factor model, the correlations between the within components of the latent factors were: isolation-engagement = -0.29 ($SE = 0.06$, $p < 0.001$); isolation-performance = -0.06 ($SE = 0.05$, $p = 0.268$); engagement-performance = 0.54 ($SE = 0.08$, $p < 0.001$). The correlations between the between components of the latent factors were: isolation-engagement = -0.09 ($SE = 0.12$, $p = 0.494$); isolation-performance = -0.13 ($SE = 0.09$, $p = 0.135$); engagement-performance = 0.65 ($SE = 0.10$, $p < 0.001$). These results supported the discriminant validity of the aforementioned scales.

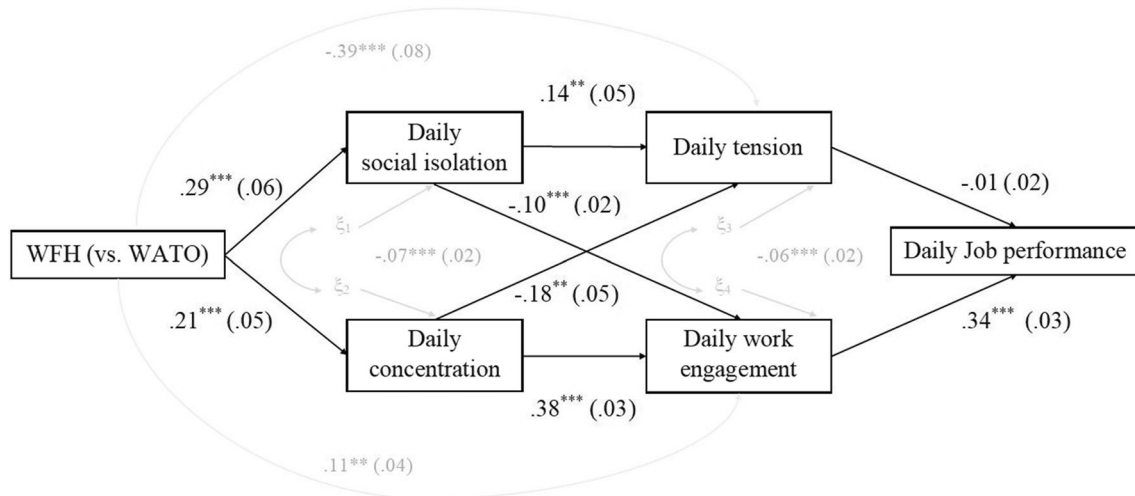
Data Analysis

Because our data were nested (daily data nested within persons), we used multilevel Structural Equation Modeling (SEM) as implemented in Mplus (Muthén & Muthén, 2017) to test our hypotheses. Considering that these hypotheses focused on relationships at the day level, we centered within-person all the model mediators (that is, daily isolation, job concentration, work engagement, and tension). Thus, scores on these variables represented participants’ day-level deviations from the corresponding participant mean across the eight considered days. Since our model only includes relationships at the day (within-person) level, to assess model fit we used the SRMR computed for the within part of our model (SRMR-within). To test the hypothesized indirect effects, we used the Monte Carlo method as implemented in RMediation (Tofghi & MacKinnon, 2011).

Table 1 Means, standard deviations, and correlations between the study variables

Variable	Mean	SD	1	2	3	4	5
1. WFH (vs. WATO)	0.33	0.47					
2. Daily social isolation	2.24	1.25	.13**				
3. Daily job concentration	5.36	0.99	.09**	-.18**			
4. Daily work engagement	5.07	0.88	.13**	-.17**	.63**		
5. Daily tension	3.44	1.56	-.18**	.07	-.17**	-.27**	
6. Daily job performance	5.28	0.70	.10**	-.12**	.48**	.54**	-.23**

WFH (vs. WATO), working from home (vs. working at the office). Coding: 0. WATO; 1. WFH. ** $p < .01$

**Fig. 2** Unstandardized parameter estimates for the research model. Note. WFH, working from home; WATO, working at the office. ** $p < .01$; * $p < .05$

Results

Table 1 reports means, SDs, and correlations between the study variables at the day level.

Intraclass Correlation Coefficients

We computed the Intraclass Correlation Coefficient (ICC(1)) to estimate the proportion of variance in the mediators and the outcome variable at the between-person and within-person levels. The ICC(1) values obtained at the between-person level were as follows: isolation: 0.51, concentration: 0.36, tension: 0.46, work engagement: 0.52, performance: 0.56. Therefore, the proportions of variance at the within-person (day) level ($1 - \text{ICC}(1)$) were as follows: 0.49, 0.64, 0.54, 0.48, and 0.44, respectively. These last values indicated that there was a substantive proportion of the aforementioned variables' variance at the within-person level.

Hypotheses Testing

The hypothesized model showed a good fit to the data (SRMR-within=0.036). In Fig. 2, we show the unstandardized regression parameter estimates obtained for the relationships included in our research model. WFH (vs. WATO) showed a positive relationship with daily social isolation ($B = 0.29$, $p < 0.001$) and daily concentration ($B = 0.21$, $p < 0.001$). Daily social isolation showed a positive and significant relationship with daily tension ($B = 0.14$, $p < 0.01$) and a negative relationship with daily work engagement ($B = -0.10$, $p < 0.001$). Daily concentration showed a positive relationship with daily work engagement ($B = 0.38$, $p < 0.001$) and a negative relationship with daily tension ($B = -0.18$, $p < 0.01$). The latter variable was not related to daily job performance ($B = -0.01$, $p = 0.422$), and daily work engagement was positively related to daily job performance ($B = 0.34$, $p < 0.001$). Finally, the direct relationships between WFH (vs. WATO), on the one hand, and daily tension and daily work engagement, on the other side, that

Table 2 Point estimates and 95% confidence intervals for the hypothesized indirect effects

Hypothesized indirect effects	Estimate	SE	95% CI
<i>Hypothesis 1</i> WFH (vs. WATO) → Daily social isolation → Daily tension → Daily job performance	.00	.001	–.002, .001
<i>Hypothesis 2</i> WFH (vs. WATO) → Daily concentration → Daily work engagement → Daily job performance	.03	.007	.015, .041
<i>Hypothesis 3</i> WFH (vs. WATO) → Daily social isolation → Daily work engagement → Daily job performance	–.01	.003	–.015, –.005
<i>Hypothesis 4</i> WFH (vs. WATO) → Daily concentration → Daily tension → Daily job performance	.00	.001	–.001, .002

we controlled for were positive and statistically significant ($B = 0.39$, $p < 0.001$; $B = 0.11$, $p < 0.01$, respectively).

The unstandardized point estimates for the hypothesized indirect effects and their corresponding 95% confidence intervals (CI) are shown in Table 2. The indirect effect of WFH (vs. WATO) on daily job performance via daily social isolation and daily tension (Hypothesis 1; point estimate = 0.00, SE = 0.001) was not statistically significant because the 95% confidence interval included zero (–0.002, 0.001). Thus, Hypothesis 1 was not supported. The indirect effect of WFH (vs. WATO) on daily job performance via daily job concentration and daily work engagement (Hypothesis 2; point estimate = 0.03, SE = 0.008) was statistically significant because the 95% confidence interval excluded zero (0.015, 0.039). Therefore, Hypothesis 2 was supported. The indirect effect of WFH (vs. WATO) on daily job performance via daily social isolation and daily work engagement (Hypothesis 3; point estimate = –0.01, SE = 0.003) was statistically significant because the 95% confidence interval excluded zero (–0.015, –0.005). Therefore, Hypothesis 3 was supported. The indirect effect of WFH (vs. WATO) on daily job performance via daily concentration and daily tension (Hypothesis 4; point estimate = 0.00, SE = 0.001) was not statistically significant because the 95% confidence interval included zero (–0.001, 0.002). Thus, Hypothesis 4 was not supported.¹

To obtain an effect size measure of the statistically significant indirect effects (i.e., those involved in Hypotheses 2 and 3), and considering that WFH (vs. WATO) was a dichotomous variable, we computed the corresponding Partially

Standardized Indirect Effects (PSIE) (Preacher & Kelley, 2011; Wen & Fan, 2015). PSIE was computed as follows: $PSIE = \frac{B_{M_1X} \cdot B_{M_2M_1} \cdot B_{YM_2}}{\sigma_Y}$, where B_{M_1X} is the unstandardized structural coefficient estimating the relationship between the predictor (X: WFH vs. WATO) and the first mediator (M_1 : daily concentration/daily social isolation), $B_{M_2M_1}$ is the unstandardized structural coefficient estimating the relationship between the first mediator (M_1 : daily concentration/daily social isolation) and the second mediator (M_2 : daily work engagement), B_{YM_2} is the unstandardized structural coefficient estimating the relationship between the second mediator and the outcome variable (Y: daily performance), and σ_Y is the standard deviation of the outcome variable. Thus, PSIE is the ratio of the corresponding indirect effect (see the numerator of the formula shown above) to the standard deviation of the outcome variable (see its denominator) (Preacher & Kelley, 2011).

For the indirect effect of WFH (vs. WATO) on daily job performance via daily concentration and daily work engagement (Hypothesis 2), the PSIE equaled 0.04, which means that daily job performance indirectly increased by 0.04 SDs on days in which participants worked from home as compared with days in which they worked at the office via daily concentration and daily work engagement. For the indirect effect of WFH (vs. WATO) on daily job performance via daily social isolation and daily work engagement (Hypothesis 3), the PSIE equaled –0.01, which means that daily job performance indirectly decreased by 0.01 SDs on days in which participants worked from home as compared with days in which they worked at the office via daily social isolation and daily work engagement. As reported in Footnote 1, our model also included a statistically significant positive indirect effect involving only one mediator (WFH (vs. WATO) → Daily work engagement → Daily job performance). The corresponding PSIE equaled 0.05. Therefore, the combined PSIE involving the two observed positive indirect effects (0.04 + 0.05) equaled 0.09, and the combined PSIE involving the three indirect effects (0.04 + 0.05 – 0.01), equaled 0.08.

¹ Our model also included two additional indirect effects based on previous empirical findings, which we controlled for (see the last paragraph of the Introduction section): WFH (vs. WATO) → Daily tension → Daily job performance, and WFH (vs. WATO) → Daily work engagement → Daily job performance. The one involving daily tension as the mediator was not statistically significant (point estimate = .01, SE = .006, 95% CI = [–.015, .005]). However, the one involving daily work engagement as the mediator (WFH (vs. WATO) → Daily work engagement → Daily job performance) was statistically significant (point estimate = .04, SE = .013, 95% CI = [.015, .057]).

Discussion

The goal of our diary study was to test a mediation model to identify some of the mediators that transmit the positive and the negative influences of WFH (vs. WATO) on job performance. We found that WFH (vs. WATO) was positively related to daily concentration, which in turn was positively related to work engagement, which in turn was positively related to daily job performance. The positive indirect effect of WFH (vs. WATO) on daily job performance via daily concentration and daily work engagement was statistically significant, which supported Hypothesis 2 and the motivational process proposed by the JD-R theory. However, the negative indirect effect of WFH (vs. WATO) on daily job performance through daily social isolation and daily tension was not statistically significant. Thus, Hypothesis 1 and the involved health-impairment process based on the JD-R theory were not supported. Regarding the hypothesized cross-relationships, we found empirical support for the negative indirect effect of WFH (vs. WATO) on daily job performance via daily social isolation and daily work engagement (Hypothesis 3). However, we did not find support for the positive indirect effect of WFH (vs. WATO) on daily job performance through daily concentration and daily tension (Hypothesis 4).

Implications for Theory and Research

Our findings have a number of theoretical implications that we will now discuss. First, our results uncover two of the mechanisms through which WFH (vs. WATO) is related to job performance, offering a more fine-grained explanation about *why* and *how* the two variables are related. According to our theoretical framework and the results obtained, one of the mechanisms identified (WFH (vs. WATO) → Daily concentration → Daily work engagement → Daily job performance) suggests that WFH provides employees with a useful job resource: concentration. WFH implies a work environment with fewer interruptions and less noise, which helps employees to focus their attention on work activities (Xiao et al., 2021). Then, concentration fosters work engagement because the former is “the breeding ground for absorption” (Bakker, 2008, p. 406), a key dimension of work engagement. Moreover, job concentration helps employees to perform more efficiently, which contributes to fulfilling their need for competence, which fosters motivation states like work engagement (Ryan & Deci, 2000). Finally, work engagement improves performance because the former increases employees’ cognitive functioning, cognitive openness, behavioral flexibility, and thought and action repertoires (Fredrickson, 2001;

Reijseger et al., 2010). Thus, this mechanism helps us to understand *why* WFH (vs. WATO) is *positively* related to job performance.

The other mechanism we identified (WFH (vs. WATO) → Daily social isolation → Daily work engagement → Daily job performance) and our theoretical framework suggests that WFH implies a physical and psychological distance from colleagues that contributes to increasing their perceptions of social isolation (Fonner & Roloff, 2010; Pyöriä, 2011). Then, social isolation is negatively related to the motivational state of work engagement because it hampers the satisfaction of a primary human need: the need of relatedness, which is one of the bases of human motivation (Deci et al., 2017; Ryan & Deci, 2000). Finally, as explained above, work engagement is positively related to job performance. Thus, this second mechanism helps us to understand *why* WFH (vs. WATO) is *negatively* related to job performance.

The two identified mechanisms point out that the relationship between WFH (vs. WATO) is, as Allen and colleagues (2015) suggested, “complex, with the potential for simultaneous benefits and drawbacks” (p. 60). By providing a detailed explanation about the indirect positive and negative effects of WFH on job performance and identifying the mediators involved, our study contributes to advancing knowledge in the field and moving our discipline forward (Mathieu et al., 2008).

Second, our study shows that, when tested together, the motivational process embedded within the JD-R model was supported by our results, but the health-impairment process was not. These results may be due to the outcome we investigated: job performance. Earlier formulations of the JD-R theory suggested that the health-impairment process was conceived to explain health-related outcomes, such as health problems or sickness absenteeism, whereas the motivational process was more appropriate to explain performance-related variables (Bakker et al., 2003; Schaufeli & Bakker, 2004). Also, our results suggest that in addition to the motivational process we identified (WFH (vs. WATO) → Daily concentration → Daily work engagement → Daily job performance), there is a *disengagement process* (González-Romá et al., 2020) through which WFH leads to decreased work engagement via social isolation and to reduced job performance via social isolation and work engagement. González-Romá et al. (2020) observed a similar disengagement process at the individual and team levels when examining the relationships between role overload and organizational commitment. Future studies should replicate the results observed here to ascertain whether the disengagement process is supported in different samples.

Third, and related to the previous point, our results show that there are cross-links between the motivational and health-impairment processes embedded within the JD-R

theory. Specifically, daily concentration was negatively related to daily tension and daily social isolation was negatively related to daily work engagement. Moreover, one of the hypothesized indirect effects based on these cross-links (WFH (vs. WATO) → Daily social isolation → Daily work engagement → Daily job performance; Hypothesis 3) was statistically significant. These results are congruent with other studies (Bakker et al., 2003; Schaufeli & Bakker, 2004) and meta-analysis (Alarcon, 2011; Crawford et al., 2010; Lee & Ashforth, 1996) that reported that job resources are negatively related to employee strain, whereas job demands are negatively related to motivational states. This empirical evidence suggests that the JD-R theory should be extended to include these cross-links. Doing so will allow it to give a more comprehensive account of the relationships among the involved constructs.

Fourth, we found that the net effect of WFH on daily job performance is positive since the observed significant positive indirect effect outweighs the negative one. Thus, based on our findings, we can state that the employees of our sample performed better on the days in which they worked from home than on the days in which they worked at the office. However, to enhance the positive relationship of WFH and daily job performance, future research should investigate ways to buffer the negative impact of WFH on daily job performance through daily social isolation. For instance, future studies could consider the role of social support from managers (Toscano et al., 2022), as well as the adoption of specific tools or strategies that facilitate communication when working from home (Fonner & Roloff, 2012), as potential factors that can mitigate the harmful effects of WFH on social isolation.

Practical Implications

In addition to theoretical implications, our findings also have several practical implications. First, our results show that WFH is positively related to job performance via daily concentration and work engagement. Furthermore, they indicate that, on days in which employees work from home, their concentration, engagement, and performance are higher than on days in which they work at the office. These findings suggest that managers supervising employees with a hybrid work arrangement could facilitate work engagement and performance by creating a work environment at the office that fosters concentration on work activities. This can be done in different ways. Managers can model employee behavior by avoiding constant interruptions of others' work and working silently. They can also consider the design of the workspace at the office and ensure that this space has the appropriate physical characteristics (e.g., acoustic isolation, physical distance from others, lightning) to work with high levels of concentration.

Second, our study shows that WFH is negatively related to daily job performance because the former fosters daily social isolation, which in turn is negatively related to daily work engagement, which in turn is positively related to daily job performance. To reduce this indirect negative relationship, organizations implementing hybrid work policies should develop strategies to decrease employees' feelings of social isolation that are associated with working from home. Possible solutions include offering virtual informal social activities such as coffee breaks, providing resources for maintaining social connections, planning online meetings, and encouraging employees to connect with their colleagues and supervisors at certain times along the working day.

Limitations and Strengths

This study has several limitations that must be considered when interpreting its results. First, the data analyzed in this study come from civil servants working in a public organization (a municipality). This limits the generalizability of our findings to other types of organizations. However, we want to highlight that our sample was composed of employees with varied occupations (e.g., administrative staff, engineers, and educators), which suggests that our findings should not be restricted to a specific occupation. Notwithstanding, future studies should aim to replicate our findings in different organizations and business sectors. Second, the observed response rate (46.2%) was lower than desired. Several factors might have contributed to it. The data collection was carried out during a period when the effects of the pandemic on the involved municipality were still significant. This likely increased the workload for many employees, limiting their availability to respond to the questionnaire over the eight consecutive days under study. Moreover, the data collection coincided with two local holidays, which might have affected employees' commitment to participate in the study every day. For example, some employees might have extended these holidays by taking extra days off, so that they had a full week out of work. Finally, employees did not receive any incentive for participating in the study, which probably did not encourage participation. However, the percentage of completed questionnaires under WFH and WATO conditions (33.4% and 67.6%, respectively) was in line with the percentage of weekly working time that employees could spend (according to the characteristics of the hybrid work plan implemented by the municipality) working from home (20–40%) and working at the office (80%–60%). This suggests that missing data did not disproportionately affect days when employees worked from home versus days when they worked at the office.

Third, the data analyzed were cross-sectional in nature. Therefore, causal relationships between the variables included in our mediation model cannot be assumed. Fourth,

daily tension and daily job concentration were measured by single items. Although this practice is not rare in diary studies because it contributes to reducing questionnaire length and participants' tiredness and attrition (Ohly et al., 2010; Song et al., 2022; Van Hooff et al., 2007), it has been widely accepted that single-item measures may put at risk the reliability and validity of participants' scores. However, Mathews et al. (2022) have recently shown that single-item measures can have good psychometric properties. Similarly, Song et al. (2022) have recently "demonstrated that single-item measures can exhibit adequate concurrent and predictive validity, comparable to multiple-item measures". In fact, Song et al.'s results supported the use of single items in diary studies. In this regard, our single-item measures showed good psychometric properties (as reported in the Method section). Moreover, they generally showed the expected correlations with other investigated variables. For instance, daily job concentration was positively correlated with daily work engagement ($r=0.63$, $p<0.01$), and daily tension was negatively correlated with daily job performance ($r=-0.23$, $p<0.01$). These results support the validity of our single-item measures.

Fifth, job performance data were collected from the surveyed employees. Unfortunately, given the diary nature of our study and the constraints imposed by the involved organization, collecting performance data from other sources (e.g., supervisors) was not possible. However, as we showed in the Measures section, there is empirical evidence supporting the validity of the measure of self-reported job performance that we used (Pettit et al.'s (1997) scale). Sixth, on a related note, all the study variables were collected from the same source (employees). This may have inflated their correlations due to common-method variance. However, the fact that some correlations between the study variables were close to zero (e.g., daily social isolation and tension, $r=0.07$, $p>0.05$) suggested that this was not a serious concern in our study (Spector, 2006). Notwithstanding, future studies could try to replicate our findings using other sources of data. For instance, gathering job performance data for employees' supervisors.

Seventh, the reference time point of our tension measure ("At this moment") was different from the reference time point of the other study variables ("Today"). This misalignment might have attenuated the relationships between daily tension and its antecedents and consequences in our research model. However, the study by Keren et al. (2021) provided empirical evidence for the assumption that when people report on their momentary mood at a given moment, "they integrate over the history of their experiences" (p. 1). This suggests that when our respondents reported their tension level at the end of the working day, they likely integrated previous affective experiences they had over the working day. The results of the study by Parkinson et al. (1995) seem

to support this idea. They investigated the relationships between ratings of current momentary affect (i.e., affect felt at a given moment) and summary reports of affect over longer time periods (e.g., a day). They found "a reasonable degree of correspondence between the two kinds of measure" (p. 337). Finally, the fact that all the variables were measured at the same time point (end of the working day) points out that we collected and modeled respondents' synchronous responses on the study variables at that time point. Thus, for all these reasons, we think that the aforementioned misalignment should not have had an important influence on the observed relationships.

Eight, the effect sizes associated with the statistically significant indirect effects (WFH (vs. WATO) → daily concentration → daily work engagement → daily job performance; and WFH (vs. WATO) → daily social isolation → daily work engagement → daily job performance) were small (with PSIEs of 0.04 and -0.01 , respectively). The combined PSIE of WFH (vs. WATO) on daily job performance considering these two indirect effects and the one via daily work engagement only (WFH (vs. WATO) → daily work engagement → daily job performance, PSIE = 0.05) was also small (0.08). However, considering that these effect sizes are computed on a daily basis, if we assumed that they might accumulate day after day over the course of a work week, the total effect size by the end of the week, taking into account all the days employees worked from home could be larger. Moreover, sometimes even "small effect sizes [can] tell a big story" (Cortina & Landis, 2009, p. 287). In our case, the complexity of the indirect relationships investigated (involving different mediators) and the importance of the outcome variable (job performance has often been considered as the ultimate criterion in our science, Tay et al., 2023) suggest that the observed effect sizes are non-trivial.

Finally, although job concentration (a job resource) and work engagement (a motivational state) are two distinct constructs, the item we used to measure the former ("Today, I had total concentration while working"; Nohe et al., 2014) shows some overlap with the item of the work engagement scale (UWES-3) that covers the absorption dimension ("I felt immersed in my job"). To establish to what extent this affected our results, we re-estimated the relationships included in our model operationalizing work engagement using only the items that cover the dimensions of vigor and dedication. Although there were small differences under the two conditions (see the online supplemental materials), all the parameters involving work engagement maintained their statistical significance, and the conclusions of our study were not affected. Based on these results, we decided to keep the original operationalization of work engagement by means of the UWES-3 scale for the sake of consistency with its definition (which explicitly mentions the three dimensions of vigor, dedication, and absorption).

Despite these limitations, our study also has some strengths. First, by collecting data daily, we were able to disentangle the relationship of WFH vs. WATO with the investigated mediators and outcome on a daily basis. This design offers a more nuanced approach to understand the mechanisms that are activated on the days in which employees work from home. Second, our hypotheses were based on solid theories in our field (the JD-R model, Bakker & Demerouti, 2017; Demerouti et al., 2001); and the Self-Determination Theory (Deci et al., 2017; Ryan & Deci, 2000), and related empirical research (e.g., Alarcon, 2011; Crawford et al., 2010; Lee & Ashforth, 1996). This framework suggests that the results obtained after testing the study hypotheses have a strong foundation. Third, by testing a parallel mediation model, we were able to show that WFH does have both a positive and an indirect relationship with daily job performance through different mediation paths. This allowed us to improve our understanding about complex relationships.

Conclusion

We found that the relationship between WFH (vs. WATO) and daily job performance is complex and has two sides. On the one hand, WFH is positively related to daily job performance through a motivational process that involves daily concentration and daily work engagement as serial mediators. On the other side, WFH is negatively related to daily job performance through a disengagement process that involves daily social isolation and daily work engagement as serial mediators. Overall, our study improves our understanding about *how* and *why* WFH is related to job performance and helps to disentangle this complex relationship.

Appendix

Items used in the current study.

Working from home vs. Working at the office (WFH vs. WATO)

Today, have you worked ...?

0. At the office.

1. From home.

Somewhere else. Please specify.

Daily social isolation

Today, during work...

I felt isolated.

I missed the emotional support of coworkers.

I missed informal interaction with others.

I missed face-to-face contact with coworkers.

Response scale: 1. “Not at all” to 7. “Very much”.

Daily tension

At this moment, I feel:

Relaxed 1 ... 2 ... 3 ... 4 ... 5 ... 6 ... 7 Tense.

Daily job concentration

Today, ...

I had total concentration while working.

Response scale: 1. “Not at all” to 7. “Very much”.

Daily work engagement

Today, during work...

I felt bursting with energy in my job.

I felt enthusiastic in my job.

I felt immersed in my job.

Response scale: 1. “Not at all” to 7. “Very much”.

Daily job performance

Today, ...

How do you evaluate the quality of your performance?

How do you evaluate the quantity of your performance?

How do you think your supervisor would evaluate the quality of your performance?

How do you think your supervisor would evaluate the quantity of your performance?

Response scale: 1. “Very poor” to 7. “Excellent”.

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Declarations

Competing interests The authors declare no competing interests.

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